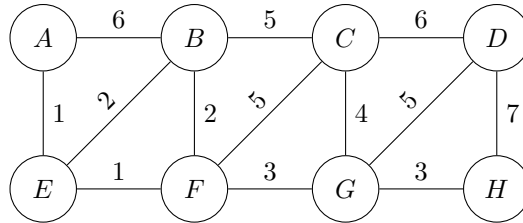
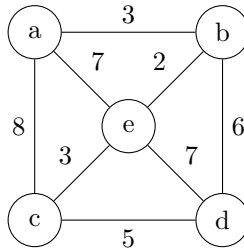


Weekly Assignment 5: Greedy Algorithms

1. (15 points) Consider the following graph:



- (a) What is the cost of its minimum spanning tree?
 - (b) How many minimum spanning trees does it have?
 - (c) Run Kruskal's algorithm on the graph, using A as initial vertex. For each iteration, indicate which edge is added to the spanning tree.
2. (15 points) We consider the following graph.



- (a) Run Prim's algorithm on the graph, using a as start vertex. For each iteration, indicate: the vertex added to the spanning tree, the priority queue for the remaining vertices and, for each vertex, predecessor in the spanning tree and key.
 - (b) Show that we can use Prim's algorithm to sort an array of size n .
3. (15 points) Consider the following definition.

Definition 1. Given a connected graph $G = (V, E)$ with edge costs c_e , a **Maximum Spanning Tree (MaxST)** is a subset of the edges $T \subseteq E$ such that T is a spanning tree whose sum of edge costs is **maximized**.

Give an algorithm to find a MaxST of a graph and discuss its complexity.

4. (25 points) Suppose that $G = (V, E)$ is a weighted connected graph and $T = (V, A)$ is a spanning tree of G . Prove the following statements or give a counterexample:

- (a) If a graph G has more than $|V| - 1$ edges, and there is always a unique heaviest edge, then this edge cannot be a part of a minimum spanning tree.
- (b) The shortest-path tree computed by Dijkstra's algorithm is always necessarily an MST.
- (c) If at least one edge of minimum weight $e \in E$ is in T then T is a MST.

5. (30 points)

We say that a sequence S' is a *subsequence* of S if there is a way to delete zero or more elements from S so that the remaining elements, in order, are equal to the sequence S' .

Give a greedy algorithm that takes two sequences — S' of length m and S of length n , each possibly containing an element more than once — and decides in time $O(n)$ whether S' is a subsequence of S .

Discuss the correctness and complexity of your algorithm.